PREFACE TO THE ENGLISH EDITION

THE advent of modern electronic computers has led to the development of new techniques of investigation, based on new principles. By means of experiments with computers, models of complex physical systems can rapidly be studied in a manner which is relatively simple and inexpensive. A most important feature is the possibility of simulating random processes on a computer. The so-called *method of statistical trials* (or *Monte Carlo method*) is, in essence, a system of techniques which enables us to model such processes conveniently in a machine. This method can be used effectively for various degrees of simulation, ranging from cases in which actual physical systems are simulated (describing them in some language which is specific for the particular problem given) to investigations of classical mathematical problems (such as systems of linear algebraic equations).

It is interesting to note that computers have led to a novel revolution in mathematics. Whereas previously an investigation of a random process was regarded as being complete as soon as it was reduced to an analytic description, nowadays it is convenient in many cases to solve an analytic problem by reducing it to a corresponding random process and then simulating that process.

In this present book, the authors have set themselves the aim of describing systematically the fundamental concepts and techniques of the Monte Carlo method, together with its range of application. The authors are grateful to Pergamon Press and to the translator Mr. G. J. Tee, who kindly accepted all proposals for improving this book. A number of additions made by the authors have been incorporated into the translation of this book. In Chapter I, § 4 and § 5 have been enlarged considerably, whilst the new § 7 and § 8 are translated from Chapters VI and IX of another book [11] written by the same author. Also, Chapter VII contains a new § 7, written especially by V. G. Sragovich for this English edition. The authors will be most gratified if this book proves to be useful for English readers.

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